



Computational Investigation of Micromagnetorotation and Thermal Transport in Williamson Nanofluid Flow over a Stretching Sheet

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Abstract: This paper presented a computational investigation of micromagnetorotation (MMR) and thermal transport characteristics in steady two-dimensional Williamson nanofluid flow over a stretching sheet. The mathematical model incorporated the non-Newtonian behavior of the Williamson fluid together with micromagnetorotational effects, magnetic field influence, viscous dissipation, and heat transfer mechanisms. By employing appropriate similarity transformations, the governing partial differential equations were reduced to a system of coupled nonlinear ordinary differential equations (ODEs). The resulting boundary-value problem was solved numerically using a shooting technique combined with the Runge–Kutta method. The effects of key physical parameters—including the Williamson parameter, magnetic parameter, MMR parameter, micropolar parameter, Prandtl number, and Eckert number—on the velocity, microrotation, and temperature distributions were examined in detail. The numerical results revealed that increasing the Williamson parameter suppressed the fluid velocity and enhanced non-Newtonian resistance within the boundary layer. Higher magnetic field strength reduced the momentum boundary-layer thickness due to the Lorentz force, while MMR significantly altered the rotational dynamics of fluid microelements. Furthermore, thermal transport was enhanced by viscous dissipation, leading to higher temperature distributions and reduced heat transfer rates at the surface. Variations in the skin-friction coefficient and local Nusselt number were reported. The current findings provided useful insights into the design and optimization of thermal systems involving non-Newtonian nanofluids subjected to micromagnetic rotational effects.

Keywords: Williamson nanofluid; Micromagnetorotation; Heat transfer; Stretching sheet; Magnetic field; Boundary layer; Simulation

1 Introduction

Micropolar fluids [1] form a special class of fluids that exhibit microstructural effects arising from the presence of suspended particles or microscopic elements capable of independent rotation. A key characteristic of these fluids is the non-symmetric nature of their stress tensor, which distinguishes them from conventional Newtonian fluids. Micropolar fluids belong to the broader family of polar fluids, and the classical Navier–Stokes equations can be recovered as a limiting case when the microstructural effects become negligible. This theory is particularly suitable for describing complex fluids containing rigid, randomly oriented, or nearly spherical particles dispersed in a viscous medium, where particle deformation is insignificant. The pioneering theory of micropolar fluids was developed by the study [2], which extended the classical fluid mechanics framework by incorporating the rotational motion of microelements and the associated couple stresses. This extension provides a more realistic representation of many engineering and industrial fluids whose behavior cannot be adequately captured by traditional Newtonian models. Owing to its ability to account for microrotational effects while maintaining a relatively simple mathematical structure, micropolar fluid theory has been commonly applied to the testing of transport processes, heat transfer, biological flows, lubricants, polymeric suspensions, and other complex fluid systems. Consequently, it continues to serve as a key instrument for both theoretical research and practical engineering applications.

Nanofluids are a modern class of heat transfer fluids formed by dispersing nanosized solid particles into conventional base fluids to improve their thermal performance [3]. These fluids are generally regarded as two-phase colloidal suspensions, in which nanoparticles constitute the dispersed phase while the base fluid such as water,

ethylene glycol, or lubricating oils, acts as the continuous phase. The addition of nanoparticles substantially enhances the thermophysical properties of the base fluid, particularly its thermal conductivity, hence making nanofluids attractive for a wide range of thermal engineering applications. Numerous types of nanoparticles have been employed in nanofluid preparation, including metals such as copper, nickel, and aluminum; metal oxides such as alumina (Al_2O_3), titanium dioxide (TiO_2), and copper oxide (CuO); as well as carbon-based materials like graphene and carbon nanotubes [4]. Due to their extremely small size and large surface-area-to-volume ratio, these nanoparticles enhance energy transport within the fluid and improve heat transfer mechanisms. As reported by Chandrasekar et al. [5], nanofluids exhibited superior thermal conductivity, improved suspension stability, reduced clogging tendencies in microchannels, and more uniform particle dispersion compared with conventional particle-fluid mixtures. The concept of nanofluids was first introduced by Choi [6] at Argonne National Laboratory, who demonstrated that even a small concentration of nanoparticles could considerably boost the heat transfer capability of a base fluid. This pioneering work stimulated extensive research into the thermal and fluid dynamic behavior of nanofluids. Among the available mathematical models, the formulation proposed by Tiwari and Das [7] has gained general endorsement because of its simplicity and effectiveness in incorporating nanoparticle volume fraction into the governing equations. The model has been successfully applied in previous studies about nanofluid transport phenomena, including micropolar and non-Newtonian fluid flows [8] where microstructural effects and enhanced thermal characteristics coexist.

Gautam et al. [9] examined the influence of Arrhenius activation energy (ACE) and binary chemical reaction (BCR) on magnetohydrodynamic (MHD) Williamson nanofluid flow over a variable-thickness expanding sheet embedded in a Darcy–Forchheimer porous medium and reported that ACE, chemical reaction, magnetic field, and porous-medium resistance significantly affect the velocity, temperature, concentration, skin-friction, and heat/mass transfer characteristics. Mishra et al. [10, 11] explored the MHD flow of a Williamson fluid through a non-Darcy porous medium in the presence of microrotation effects. Their research examined the influence of magnetic and porous medium parameters on the flow and microrotation characteristics of the fluid. Meanwhile, Agarwal et al. [12] validated the effects of viscous dissipation and Ohmic heating on the bidirectional flow of micropolar and Williamson fluids through a porous medium over a stretching surface. Their examination, moreover, covered the influence of velocity slip and convective boundary conditions on the flow and characteristics of heat transfer. While discussing the MHD flow of a Williamson micropolar nanofluid over a nonlinear stretching sheet, Kerrouche et al. [13] utilized the Williamson fluid model to capture the non-Newtonian shear-thinning behavior commonly encountered in industrial polymer extrusion processes and assessed the effects of the magnetic field on the flow and heat transfer characteristics. Williamson nanofluid flow has appealed to researchers due to its widespread applications in engineering and industrial processes. Numerous studies have been conducted to review its flow and heat transfer characteristics under different physical conditions and geometrical configurations [14–16].

More recently, Awan and Khan [17] have attempted to understand the role of micromagnetorotation (MMR) in micropolar fluid flow over a stretching surface. The scholars developed a mathematical framework based on micropolar fluid theory and incorporated anisotropic magnetization effects to capture the interaction between magnetic fields and microstructural fluid rotation. Through appropriate similarity transformations, the governing equations were reduced to a coupled system of nonlinear differential equations, introducing several dimensionless parameters that characterize the micromagnetorotational behavior of the flow.

Motivated by previous literature, the present study extended the concept of MMR to a Williamson nanofluid within a micropolar framework. The objective is to interpret how MMR influences the flow and thermal transport characteristics of a non-Newtonian nanofluid over a stretching sheet. To the best of the authors' knowledge, the combined effects of Williamson rheology, nanoparticle-enhanced heat transfer, and MMR have not yet been investigated before. To fill a vital research gap in the existing literature, this paper specifically analyzed the influence of MMR on the velocity, microrotation, and temperature fields of a two-dimensional incompressible Williamson nanofluid. Special attention was given to the thermal boundary-layer behavior and the associated heat transfer characteristics under varying physical parameters. It is expected that the results could provide useful insights into the development of advanced thermomagnetic systems and heat transfer technologies featuring complex non-Newtonian fluids.

The inclusion of MMR in the present Williamson nanofluid model was intended to represent the rotational motion of magnetic particles when the fluid was subjected to an external magnetic field. While conventional MHD nanofluid models mainly focused on the effect of the magnetic field on the bulk fluid motion, the present formulation additionally considered the rotation of magnetic microelements caused by magnetic torque. This phenomenon is particularly relevant in applications involving ferrofluids and magnetic nanofluids, such as biomedical devices, targeted drug delivery systems, microfluidic technologies, magnetic cooling equipment, and advanced heat-transfer systems. The existence of MMR could influence both the flow and thermal fields by modifying momentum transport, changing boundary-layer characteristics, and affecting heat transfer rates. Therefore, incorporating this mechanism offered a comprehensive description of magnetically responsive non-Newtonian nanofluids and allowed

a systemic understanding of their behavior in practical engineering applications where particle rotation is prevailing. The remainder of this paper is organized as follows. Section 2 presents the mathematical formulation of the micromagnetorotative Williamson nanofluid model. Section 3 describes the numerical procedure taken to solve the governing equations. Section 4 discusses the numerical results and their physical implications, while Section 5 summarizes the main findings and leads to a conclusion.

2 Micromagnetorotation Heat Transfer Model and Flow Description

We considered a two-dimensional and incompressible micropolar Williamson nanofluid flowing over a stretching sheet under the influence of an applied magnetic field $H = (0, 0, H_0)$, as depicted in Figure 1. The constitutive relationship for an incompressible and non-Newtonian Williamson fluid was mathematically expressed via the Cauchy stress tensor $\mathbf{S}$ as follows [18]:

$$\mathbf{S} = -p\mathbf{I} + \boldsymbol{\tau} \quad (1)$$

where, p represents the fluid pressure, $\mathbf{I}$ is the identity tensor, and $\boldsymbol{\tau}$ denotes the extra (or dynamic shear) stress tensor defined by the study [19]:

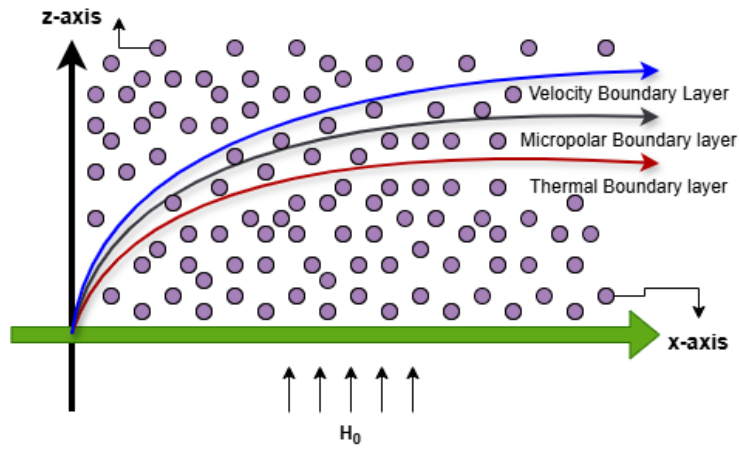


Figure 1. Geometry of the problem

$$\boldsymbol{\tau} = \left[\mu_\infty + \frac{\mu_0 - \mu_\infty}{1 - \Gamma \dot{\gamma}} \right] \mathbf{A}_1 \quad (2)$$

Here, μ_0 signifies the limiting dynamic viscosity at zero shear rate, μ_∞ represents the limiting dynamic viscosity at infinite shear rate, Γ is the Williamson material time constant, and $\mathbf{A}_1$ is the first Rivlin–Ericksen tensor defined using the velocity vector field $\mathbf{V}$ as:

$$\mathbf{A}_1 = \nabla \mathbf{V} + (\nabla \mathbf{V})^T \quad (3)$$

The scalar quantity $\dot{\gamma}$ represents the second invariant of the strain rate tensor, which is explicitly computed via the trace of $\mathbf{A}_1^2$ as:

$$\dot{\gamma} = \sqrt{\frac{1}{2} \text{tr}(\mathbf{A}_1^2)} \quad (4)$$

For boundary layer flows exhibiting pseudoplastic behavior, it is conventionally assumed that the infinite shear rate viscosity is negligible ($\mu_\infty \rightarrow 0$). Under the mathematical constraint that $\Gamma \dot{\gamma} < 1$, the application of a binomial series expansion simplifies the extra stress tensor formulation to:

$$\boldsymbol{\tau} = \mu_0 [1 + \Gamma \dot{\gamma}] \mathbf{A}_1 \quad (5)$$

When scaled to a two-dimensional boundary layer framework ($u(x, z)$), the predominant shear stress component τ_{xz} simplifies cleanly to the study [20]:

$$\tau_{xz} = \mu_{nf} \left[\frac{\partial u}{\partial z} + \frac{\sqrt{2}}{2} \Gamma \left(\frac{\partial u}{\partial z} \right)^2 \right] \quad (6)$$

The corresponding velocity, microrotation, and temperature distributions are expressed as follows:

$$\begin{aligned}\vec{V} &= (u(x, z), 0, v(x, z)), \\ \mathbf{W} &= (0, w(x, z), 0), \quad T = T(x, z)\end{aligned}$$

The governing equations are described as the study [21]:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} = 0 \quad (7)$$

$$\begin{aligned}\rho_{\text{nf}} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial z} \right) &= -\frac{\partial p}{\partial x} + J_y B_z + \eta_{\text{nf}} \left(\frac{\partial^2 u}{\partial z^2} + \sqrt{2} \Gamma \left(\frac{\partial u}{\partial z} \right) \frac{\partial^2 u}{\partial z^2} \right) + M_x \frac{\partial H_x}{\partial x} + M_z \frac{\partial H_z}{\partial x} \\ &\quad - 2\eta_1 \left(\frac{\partial w}{\partial z} - \frac{1}{2} \left(\frac{\partial^2 u}{\partial z^2} - \frac{\partial u}{\partial z \partial x} \right) \right)\end{aligned} \quad (8)$$

$$u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial z} = \gamma \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial z^2} \right) - 2\eta_1 \left(2w + \frac{\partial u}{\partial z} - \frac{\partial u}{\partial x} \right) + M_z H_x - M_x H_z \quad (9)$$

and

$$\begin{aligned}(\rho c_p)_{\text{nf}} \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial z} \right) &= \kappa_{\text{nf}} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) + \eta_2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} \right) \\ &\quad + \eta_{\text{nf}} \left[\left(\frac{\partial u}{\partial z} \right)^2 + \frac{\sqrt{2}}{3} \Gamma \left(\frac{\partial u}{\partial z} \right)^3 \right]\end{aligned} \quad (10)$$

$$\nabla \cdot \vec{B} = 0 \quad (11)$$

$$\vec{B} = \mu_0 \vec{H} + \vec{M} \quad (12)$$

$$\vec{M} = \frac{M_0(\mathbf{I} - \tau \mathbf{W} \cdot \epsilon) \cdot \vec{H}}{\bar{H}} \quad (13)$$

where, in Eqs. (7)–(13), $\vec{V} = (u, 0, v)$, $\vec{H}$, P , $\vec{w}$, $\mathbf{W} = (0, w, 0)$, $\mathbf{M} = (M_x, M_y, M_z)$, M_0 , T and B are the velocity field vector, the applied magnetic field, pressure field, angular velocity, microrotation velocity, magnetization vector, magnetization strength, temperature field, and magnetic induction vector, respectively. Furthermore, η_{nf} , η_1 , and γ are shear viscosity, vortex viscosity, and angular viscosity, respectively. The term $\vec{M} \times \vec{H}$ is due to MMR.

The nanofluid constants are given by the study [22]:

$$\rho_{\text{nf}} = (1 - \phi)\rho_f + \phi\rho_s \quad (14)$$

$$\mu_{\text{nf}} = \frac{\mu_f}{(1 - \phi)^{2.5}} \quad (15)$$

$$(\rho c_p)_{\text{nf}} = (1 - \phi)(\rho c_p)_f + \phi(\rho c_p)_s \quad (16)$$

$$\frac{\kappa_{\text{nf}}}{\kappa_f} = \frac{(\kappa_s + 2\kappa_f) - 2\phi(\kappa_f - \kappa_s)}{(\kappa_s + 2\kappa_f) + \phi(\kappa_f - \kappa_s)} \quad (17)$$

$$\sigma_{\text{nf}} = \sigma_f \left[1 + \frac{3(\sigma - 1)\phi}{(\sigma + 2) - (\sigma - 1)\phi} \right] \quad (18)$$

$$\sigma = \frac{\sigma_s}{\sigma_f} \quad (19)$$

where, in Eqs. (14)–(19), ϕ , ρ_f , ρ_s , κ_f , κ_s , σ_f , σ_s and $(c_p)_f$, $(c_p)_s$ are volume fraction of nanoparticles, densities, thermal conductivities, electrical conductivities, and specific heat capacities of the base fluid, and solid nanoparticles, respectively.

Subject to the boundary conditions:

$$\begin{aligned} u = U_x, \quad v = 0, \quad W = W_0 \frac{\partial u}{\partial z}, \quad T = T_w \quad \text{at } z = 0 \\ u \rightarrow 0, \quad W \rightarrow 0, \quad T \rightarrow T_\infty \quad \text{as } z \rightarrow \infty \end{aligned} \quad (20)$$

Ampère-Maxwell's Law is simplified for an electrically conducting fluid undergoing motion within a magnetic field, specifically in the context of MHD.

$$\nabla \times H = \sigma_{nf}(E + v \times B) \quad (21)$$

Eq. (21) is expressed as follows:

$$(j_x, j_y, j_z) = (0, \sigma_{nf}(vB_x - uB_z), 0) \quad (22)$$

Here, we established an initial boundary value problem based on the micropolar continuum description outlined in the preceding section, with a focus on MMR. In doing so, we proposed the following assumptions regarding the flow. The derivation of the constitutive relation for magnetization n is depicted in Eq. (13).

$$(M_x, M_y, M_z) = (\tau M_0 W, 0, M_0) \quad (23)$$

The magnetic induction vector B is expressed as:

$$(B_x, B_y, B_z) = (\tau M_0 W, 0, \mu_0 H_0 + M_0) \quad (24)$$

Eqs. (7)–(10) imply that:

$$\begin{aligned} u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial z} = \left(\frac{\mu_{nf}}{\rho_{nf}} + \frac{k}{\rho_{nf}} \right) \frac{\partial^2 u}{\partial z^2} + \frac{\mu_{nf}}{\rho_{nf}} \sqrt{2} \Gamma \left(\frac{\partial u}{\partial z} \right) \frac{\partial^2 u}{\partial z^2} + \frac{\kappa}{\rho} \frac{\partial w}{\partial z} \\ - \frac{\sigma}{\rho} [(\mu_0 H_0 + M_0) \tau M_0 v w + (\mu_0 H_0 + M_0)^2 u] \end{aligned} \quad (25)$$

$$u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial z} = \frac{\gamma^*}{\rho_{nf} j} \frac{\partial^2 w}{\partial z^2} - \frac{\kappa}{\rho_{nf} j} \left(2w + \frac{\partial u}{\partial z} \right) - \frac{\tau M_0 H_0}{\rho_{nf} c} w \quad (26)$$

and

$$\begin{aligned} u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial z} = \frac{\kappa_{nf}}{(\rho c_p)_{nf}} \frac{\partial^2 T}{\partial z^2} + \frac{\kappa_{nf}}{(\rho c_p)_{nf}} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} \right) \\ + \frac{\mu_{nf}}{(\rho c_p)_{nf}} \left[\left(\frac{\partial u}{\partial z} \right)^2 + \frac{\sqrt{2}}{3} \Gamma \left(\frac{\partial u}{\partial z} \right)^3 \right] \end{aligned} \quad (27)$$

Similarity transformations:

$$\begin{aligned} \xi = z \sqrt{\frac{U_\infty}{\nu x}}, \quad \psi = U_\infty \sqrt{\frac{\nu x}{U_\infty}} f(\xi), \quad u = U_\infty f'(\xi), \quad v = \frac{1}{2} \sqrt{\frac{U_\infty \nu}{x}} (\xi f'(\xi) - f(\xi)) \\ w = U_\infty \sqrt{\frac{U_\infty}{\nu x}} R(\xi), \quad \theta(\xi) = \frac{T - T_\infty}{T_w - T_\infty} \end{aligned} \quad (28)$$

Eqs. (25)–(27) are transformed into the following nonlinear coupled ordinary differential equations (ODEs):

$$(1 + K)f''' + \lambda f'' f''' + K R' - \frac{A}{2} (\xi f' - f) R + M f' + \frac{f f''}{2} = 0 \quad (29)$$

$$\left(1 + \frac{K}{2} \right) R'' - \beta (2R + f'') - w w^* R + \frac{f' R}{2} + \frac{f R'}{2} = 0 \quad (30)$$

$$\frac{1}{Pr}\theta'' + Ec \left(K f''^2 + f''^2 + \frac{1}{3}\lambda f''^3 \right) + \frac{f\theta'}{2} = 0 \quad (31)$$

Boundary conditions are:

$$\begin{aligned} f &= 0, \quad f' = 1, \quad R = 0, \quad \theta = 1 \quad \text{at } \xi = 0 \\ f' &= 0, \quad R = W_0 f''(0), \quad \theta = 0 \quad \text{as } \xi \rightarrow \infty \end{aligned} \quad (32)$$

Here, the parameters K , ξ , β , w , M , A , w^* , Pr , and Ec are dimensionless and are stated as follows:

$$\begin{aligned} K &= \frac{\kappa}{\mu_{nf}}, \quad \xi = \frac{H_0^2}{c^2 \ell}, \quad \left(1 + \frac{K}{2}\right) = \frac{\gamma^*}{\rho_{nf} j}, \quad \beta = \frac{\kappa}{\rho_{nf} j}, \quad \lambda = \Gamma \sqrt{\frac{2c^3 x^2}{\nu_{nf}}} \\ M &= \frac{\sigma_{nf} B_0^2}{\rho_{nf} c}, \quad w^* = \frac{\tau}{c}, \quad w = \frac{M_0 H_0^2}{\rho_{nf} \bar{H}}, \quad A = \frac{\sigma_{nf}}{\rho_{nf}} (\mu_0 H_0 + M_0) \tau M_0 \\ Pr &= \frac{(\mu c_p)_{nf}}{k_{nf}}, \quad Ec = \frac{U_\infty^2}{(c_p)_{nf} (T_w - T_\infty)} \end{aligned} \quad (33)$$

The parameter of physical interest is the skin friction coefficient and is defined as:

$$C_f = \frac{\tau_w}{\rho_{nf} u_w^2} \quad (34)$$

where, $u_w = cx$ denotes the characteristic velocity and τ_w refers to the wall shear stress given by:

$$\tau_w = \left[\mu_{nf} \left(\frac{\partial u}{\partial z} + \frac{\sqrt{2}}{2} \Gamma \left(\frac{\partial u}{\partial z} \right)^2 \right) + k_{nf} \left(\frac{\partial u}{\partial z} + W \right) \right]_{z=0} \quad (35)$$

Substituting Eq. (28) into Eq. (35) gives the following expression for the skin-friction coefficient in Eq. (34):

$$C_f \sqrt{Re_x} = (1 + K) f''(0) + \frac{\lambda}{2} [f''(0)]^2 + K R(0) \quad (36)$$

The local Nusselt number Nu_x , which characterizes the heat transfer rate at the solid boundary, is defined as:

$$Nu_x = \frac{x q_w}{\kappa_f (T_w - T_\infty)} \quad (37)$$

where, q_w represents the surface heat flux computed via Fourier's law of conduction:

$$q_w = -\kappa_{nf} \left(\frac{\partial T}{\partial z} \right)_{z=0} \quad (38)$$

By substituting the similarity transformations into the heat flux formulation, the dimensionless local Nusselt number reduces to the following elegant form:

$$(Re_x)^{-1/2} Nu_x = -\frac{\kappa_{nf}}{\kappa_f} \theta'(0) \quad (39)$$

where, $Re_x = \frac{u_w x}{\nu_f}$ is the local Reynolds number, with u_w denoting the characteristic wall stretching velocity, x being the streamwise coordinate, and ν_f representing the kinematic viscosity of the base fluid.

3 Numerical Method of Solution

3.1 Shooting Method

To solve the nonlinear and coupled ODEs governed by the system of Eqs. (29)–(31) subject to the boundary conditions in Eq. (32), a dual numerical approach was implemented to ensure maximum precision and verification. The primary solutions were computed utilizing the classical shooting method integrated with a Runge–Kutta Fehlberg (RKF-45) order scheme. To validate the reliability of these trajectories, the results were cross-examined and bolstered using the built-in MATLAB boundary value problem solver, bvp4c [23]. The shooting scheme served as a highly versatile and dependable tool for deciphering boundary layer problems, thus demonstrating exceptional stability when handling stiff differential frameworks. For these simulations, selecting an appropriate domain infinity (ξ_∞) that accurately encapsulated the physical boundary layer thickness was essential. Because the convergence and

stability of both algorithms were highly sensitive to the initial mesh discretization and guessing profiles, systematic grid-independence refinements were carried out. All numerical computing architectures were executed within the MATLAB environment. To facilitate the numerical execution via the shooting scheme, the governing higher-order system was gradually reduced to a set of simultaneous first-order ODEs. By defining the state variables as:

$$\begin{cases} y_1' = y_2 & y_1(0) = 0 \\ y_2' = y_3 & y_2(0) = 1 \\ y_3' = \left(\frac{1}{1 + K + \lambda y_3} \right) \left[\frac{A}{2} (\xi y_2 - y_1) y_4 - M y_2 - \frac{y_1 y_3}{2} - K y_5 \right] & y_3(0) = r \\ y_4' = y_5 & y_4(0) = 0 \\ y_5' = \left(\frac{2}{2 + k} \right) \left[\beta_1 (2 y_4 + y_3) + \omega \omega^* y_4 - \frac{y_2 y_4}{2} - \frac{y_1 y_5}{2} \right] & y_5(0) = s \\ y_6' = y_7 & y_6(0) = 1 \\ y_7' = -Pr \left[Ec \left(K y_3^2 + y_3^2 + \frac{1}{3} \lambda y_3^3 \right) + \frac{y_1 y_7}{2} \right] & y_7(0) = t \end{cases} \quad (40)$$

The domain was split into 1,000 linear elements. This system was then solved together with bvp4c. The computational domain was chosen to be $[0, 14]$ instead of the infinite domain $[0, \infty)$; this is due to the attainment of the boundary layer development. For the validation of the results obtained through this implementation, the problem of the model was resolved by MATLAB solver. The collocation technique and a point mesh were applied to divide the integration interval into smaller intervals. To verify the accuracy of the obtained solutions, skin-friction coefficient was computed with bvp4c and shooting methods against the material parameters K , A , and β , with varying values of the MMR parameter ω^* . Table 1 displays the calculated solutions in relation to the skin-friction coefficient. These calculations manifested that the skin-friction coefficient values are in strong agreement with the values from the shooting method.

Table 1. Variation of skin-friction coefficient $C_f \sqrt{Re_x}$ with governing physical parameters in Williamson nanofluid flow over a stretching surface under magnetohydrodynamic (MHD) and microrotation effects

Parameter	K	A	β	ω^*	Skin-Friction Coefficient
Micropolar parameter K	0.0	0.01	0.5	5.5	0.036908
	0.5	—	—	—	0.082973
	1.0	—	—	—	0.129086
	1.5	—	—	—	0.175222
Magnetization parameter A	—	0.0	—	—	0.175136
	—	0.01	—	—	0.175165
	—	0.02	—	—	0.175194
	—	0.03	—	—	0.175222
Material parameter β	—	—	0.5	—	0.175204
	—	—	1.0	—	0.175222
	—	—	1.5	—	0.175235
	—	—	2.0	—	0.175244
Micromagnetorotation (MMR) parameter ω^*	—	—	—	0.0	0.175272
	—	—	—	2.5	0.175240
	—	—	—	5.5	0.175222
	—	—	—	7.5	0.175215

Note: The symbol “—” indicates that the corresponding parameter was held constant at its baseline value while the specified parameter was varied.

4 Results and Discussion

This section unveils the numerical solutions of the MMR initial boundary value problem described by Eqs. (29)–(31). Numerical simulations were conducted for varying values of the material parameters to calculate the translational and rotational velocity profiles as well as temperature profiles in micropolar cases. A complete discussion on Table 1 has been done to reveal the impact of dimensionless parameters K , A , β , and ω^* on the skin friction coefficient. The skin-friction coefficient increases with K and β , while it decreases slightly with increasing

ω^* . Table 2 further presents the variation of the local Nusselt number with the micropolar parameter K , material parameter β , MMR parameter ω^* , and Prandtl number Pr .

Table 2. Variation of Nusselt number Nu with governing physical parameters in Williamson nanofluid flow over a stretching surface under magnetohydrodynamic (MHD) and thermal effects

Parameter	K	β	ω^*	Pr	Nusselt Number
	0.0	0.01	0.5	5.5	0.109323
Micropolar parameter K	0.0	—	—	—	0.109629
	1.0	—	—	—	0.109803
	1.5	—	—	—	0.109918
	—	0.5	—	—	0.109892
Material parameter β	—	1.0	—	—	0.109918
	—	1.5	—	—	0.109935
	—	2.0	—	—	0.109948
	—	—	0.0	—	0.109990
Micromagnetorotation (MMR) parameter ω^*	—	—	2.5	—	0.109944
	—	—	5.5	—	0.109918
	—	—	7.5	—	0.109907
	—	—	—	2.0	0.322280
Prandtl number Pr	—	—	—	4.0	0.217377
	—	—	—	6.0	0.127859
	—	—	—	8.0	0.127860

Note: The symbol “—” indicates that the corresponding parameter was held constant at its baseline value while the specified parameter was varied.

Figure 2 illustrates the effect of the magnetization parameter A on the translational velocity profile. Physically, the magnetization parameter represents the strength of magnetic interactions within the conducting Williamson micropolar nanofluid. As the magnetization parameter increases, the translational velocity increases slightly throughout the boundary layer, indicating that the magnetic interactions promote the fluid motion under the present flow conditions. This behavior may be attributed to the modification of momentum transport induced by the magnetization effects, which facilitates the movement of the conducting fluid. Consequently, the momentum boundary-layer thickness increases slightly with increasing values of A . These results demonstrate that the magnetization parameter plays an important role in controlling the fluid motion and may be utilized to regulate the flow characteristics of electrically conducting Williamson micropolar nanofluids. Figure 3 illustrates the influence of the micropolar parameter K on the translational velocity profile. Physically, the micropolar parameter characterizes the coupling between the linear and angular motions of the fluid microelements. As the micropolar parameter increases, the translational velocity increases throughout the boundary layer, indicating that the enhanced coupling between the fluid velocity and microrotation promotes momentum transport under the present flow conditions. Consequently, the momentum boundary-layer thickness increases slightly with increasing values of K . The stronger interaction between the translational and rotational motions modifies the velocity distribution and facilitates the fluid motion. These results demonstrate that the micropolar parameter significantly influences the flow behavior by altering the momentum transfer within the Williamson micropolar nanofluid.

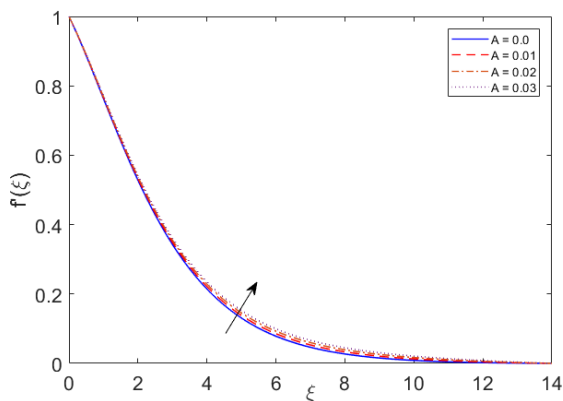


Figure 2. Translational velocity for varying A

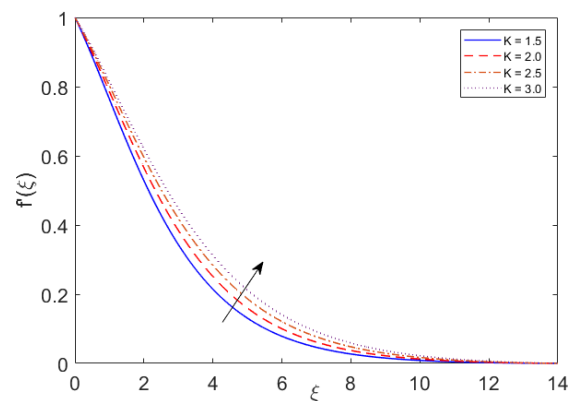


Figure 3. Translational velocity for varying K

The material parameter β significantly influenced both the flow and thermal characteristics of the Williamson fluid through its effect on the fluid microstructure and rheological behavior. Physically, β represents the strength of the non-Newtonian characteristics of the fluid. An increase in β enhanced the fluid's resistance to deformation, which suppressed the fluid velocity and reduced the thickness of the momentum boundary layer. Quantitatively, increasing β decreased the peak velocity. At the same time, the reduced fluid motion weakened convective cooling and allowed more thermal energy to accumulate within the boundary layer, resulting in an increase in the temperature profile. Therefore, higher values of β suppressed momentum transport while promoting thermal diffusion within the flow domain. In Figure 4, the effect of material parameter on translational velocity profile is observed, as the profile would be high with a decline in the values of material parameter. The MMR parameter is an idea that can develop during research on materials or fluids driven by both microrotational effects and magnetic fields. In general, this property depicts the small-scale rotational effects, or microrotation, that are impacted by a magnetic field. In such systems, the translational velocity, rotational velocity, and temperature profiles are largely determined by the MMR parameter. Physically, the MMR parameter represents the rotational interaction of magnetic microelements suspended within the conducting micropolar nanofluid.

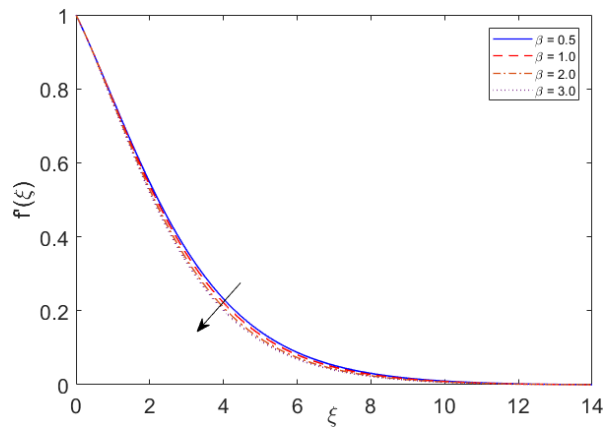


Figure 4. Translational velocity for varying β

An increase in ω^* enhanced the rotational motion of magnetic particles, which contributed additional kinetic energy to the fluid motion. This weakened the resistive effects within the boundary layer and slightly accelerated the translational flow. As a result, the momentum boundary layer thickness increased marginally as the MMR parameter increased. Figure 5 represents how MMR affects the translational velocity profile. The velocity profile evidenced an increase with incrementing the MMR parameter. Figure 6 displays the impact of A on the rotational velocity profile.

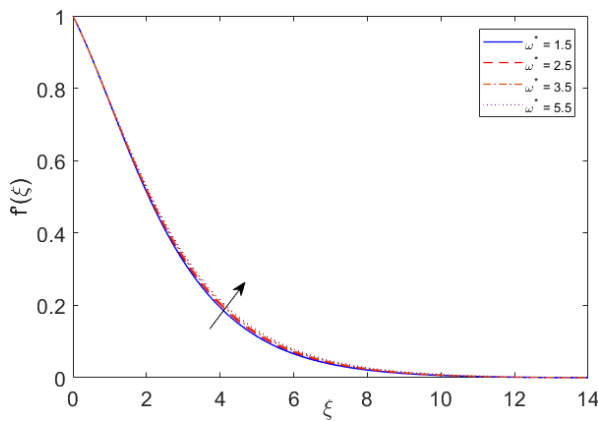


Figure 5. Translational velocity for varying ω^*

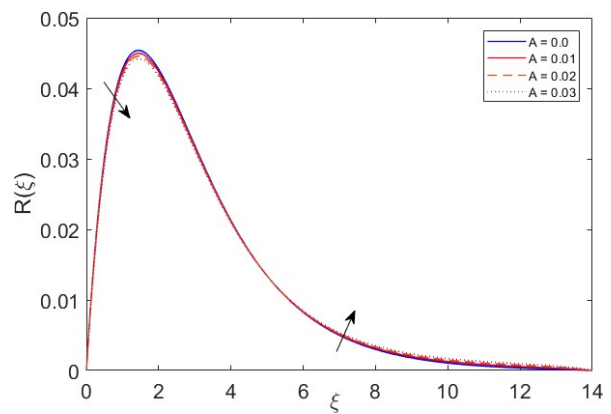


Figure 6. Rotational velocity for varying A

Substantially, the magnetizing parameter controlled the magnetic interaction strength within the fluid. Stronger magnetization increased the magnetic torque acting on the microelements of the fluid, thereby modifying the rotational motion. Near the surface, the enhanced magnetic resistance suppressed the rotational velocity, while farther from the wall the redistribution of angular momentum slightly increased the rotational profile. The effect of magnetization on boundary layer profiles was observable in a specific pattern in the vicinity of the stretching surface relative to the distant vicinity. The observed patterns in the boundary layer profiles could be affected by various

factors, such as fluid characteristics and the geometry of the flow.

The velocity profiles indicate a trend to decrease in the vicinity of the stretching surface and increase in the boundary layer region. Physically, K represents the coupling between the linear momentum and microrotation of microstructural particles suspended in the fluid. Larger values of K strengthened the interaction between the fluid particles and their spin motion, which enhanced the resistance to fluid rotation close to the surface. Consequently, the peak rotational velocity was reduced. However, farther away from the wall, the rotational profiles became slightly larger for higher K , implying a thicker rotational boundary layer and slower decay of microrotation effects in the outer region. With an increase in micropolar parameter, the translational velocity tends to decrease near the wall of sheet and high with boundary layer regions. This is so since rotational motion is resisted by increased couple stresses, which growing internal friction in the fluid. Higher couple stresses cause the fluid to become more resistant to flow, which slows down translational velocity. Figure 7 examines the effect of the micropolar parameter on rotational velocity profile. The velocity of the fluid increased in the boundary layer region and decreased near the wall with micropolar parameter. Thus, a Lorentz force decreased the velocity field. From the physics perspective, the magnetic field became stronger and the Lorentz force rose.

Figure 8 reflects the influence of the MMR parameter ω^* on the rotational velocity profile $R(\xi)$. It is observed that increasing ω^* significantly suppressed the rotational motion throughout the boundary layer, resulting in lower values of $R(\xi)$ and a thinner rotational boundary layer. Quantitatively, increasing ω^* reduced the peak rotational velocity. Physically, larger values of ω^* increased the resistance to the rotation of fluid microelements and promoted the dissipation of microrotational energy. Hence, less angular momentum was available to sustain particle rotation, causing the rotational velocity to decrease. This behavior indicates that the MMR parameter acts as a rotational damping mechanism, weakening the spin of microelements and reducing the extent of rotational motion within the fluid. Figure 9 illustrates the effect of the material parameter β on the rotational velocity profile. The results show that the rotational velocity increases with increasing values of the material parameter. Physically, the material parameter modifies the rheological characteristics of the Williamson fluid and influences the coupling between the fluid motion and the microrotation of the fluid microelements. As β increases, the rotational motion of the fluid particles is enhanced, leading to higher rotational velocity throughout the boundary layer. This enhancement is more pronounced near the stretching surface, whereas the rotational velocity gradually approaches zero away from the wall, satisfying the prescribed boundary condition. These results indicate that the material parameter plays an important role in controlling the rotational dynamics and transport characteristics of the Williamson micropolar nanofluid. Figure 10 illustrates the effect of the magnetization parameter A on the temperature profile. The results indicate that increasing the magnetization parameter reduces the temperature throughout the thermal boundary layer. Physically, stronger magnetization enhances the interaction between the applied magnetic field and the electrically conducting Williamson micropolar nanofluid. This interaction generates additional magnetic damping, which suppresses thermal energy transport and reduces the fluid temperature. Consequently, the thermal boundary layer becomes thinner, and the temperature profile decays more rapidly as A increases. These results demonstrate that the magnetization parameter significantly influences the thermal characteristics of the flow by reducing the temperature distribution within the boundary layer.

The physical behavior of temperature profile versus variations in micropolar parameter is depicted in Figure 11. Micropolar parameter K increases, the temperature profile increases, resulting in a thicker thermal boundary layer. Physically, the micropolar parameter represents the coupling between the translational motion of the fluid and the microrotation of its microelements. An increase in K strengthens this coupling, allowing more energy to be transferred from the mean flow to the rotational motion of the suspended microstructures. This interaction enhances internal friction and dissipative effects, generating additional thermal energy within the fluid. Consequently, the fluid temperature rises and the thermal boundary layer becomes thicker. The fluid's temperature increased as the intensity in micropolar parameter increased, as seen in the graph. Figure 12 demonstrates the impact of Ec on the temperature profile. Raising the values of Eckert number, the temperature profile increased due to the decrement in heat transfer rate. This behavior was physically attributed to the phenomenon of viscous dissipation, where kinetic energy was converted into thermal energy due to internal fluid friction and nanoparticle shear stresses, thereby raising the fluid's thermal state. The Prandtl number has a direct impact on the temperature in a fluid flow, especially in boundary layer flows as momentum and the simultaneous occurrence of heat transfer. Escalating values of Pr caused a distinct decline in the fluid temperature and a reduction in the thermal boundary layer thickness. Because the Prandtl number represents the ratio of momentum diffusivity to thermal diffusivity, larger values of Pr signify a reduced thermal conductivity. It follows that the heat diffusion capacity of the micropolar nanofluid weakens, restricting thermal penetration into the fluid domain and shrinking the thermal boundary layer. Figure 13 showcases the effect of the Prandtl number on the temperature distribution. From the results, it can be deduced that the fluid temperature decreased with increasing Prandtl number, hence reflecting a reduction in the fluid's thermal conductivity. As the Prandtl number rose, the diminished thermal diffusivity led to a lower temperature field within the fluid. Figure 14 depicts the influence of the material parameter on the temperature profile. It is evident that the temperature profile

decreases with increasing values of the material parameter β . Physically, the material parameter β is inversely proportional to the fluid yield stress. Smaller values of β correspond to a higher yield stress, which enhances the fluid's plastic resistance and internal shear during motion. Consequently, greater viscous dissipation occurs, generating additional thermal energy within the boundary layer and increasing the nanofluid temperature. In contrast, as β increases, the yield stress decreases and the fluid gradually approaches Newtonian behavior. This rheological transition weakens the internal frictional effects, thereby reducing viscous heat generation. As a result, the fluid temperature declines, leading to a thinner thermal boundary layer. This observed cooling trend with increasing β is in good agreement with previously reported results for non-Newtonian Williamson boundary-layer flows.

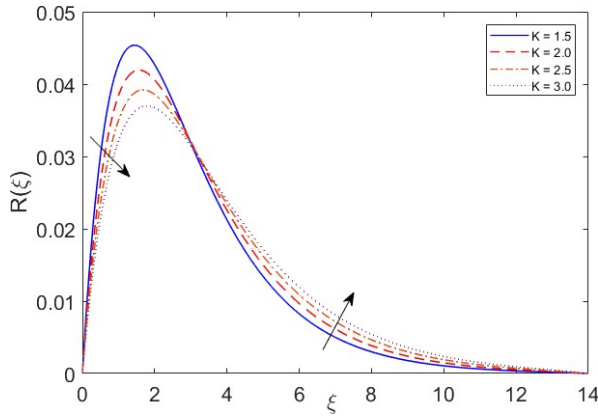


Figure 7. Rotational velocity for varying K

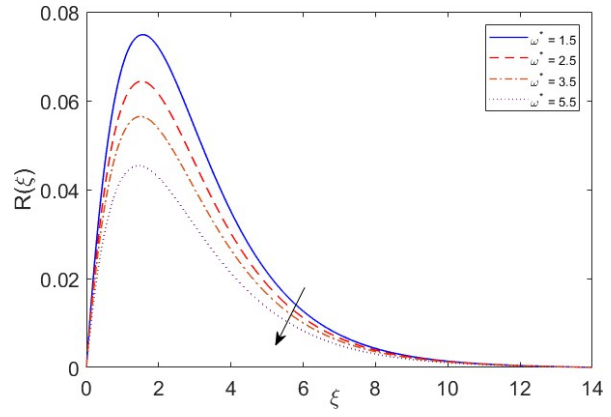


Figure 8. Rotational velocity for varying ω^*

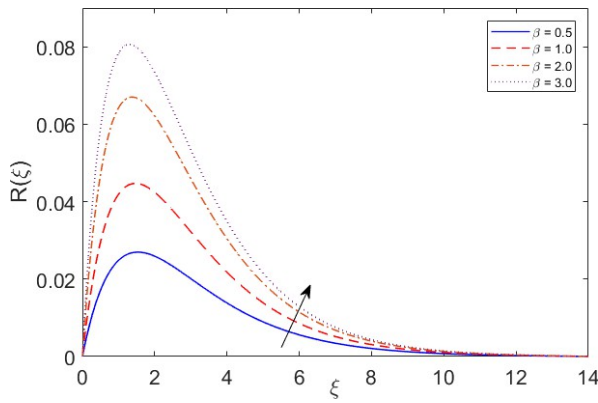


Figure 9. Rotational velocity for varying β

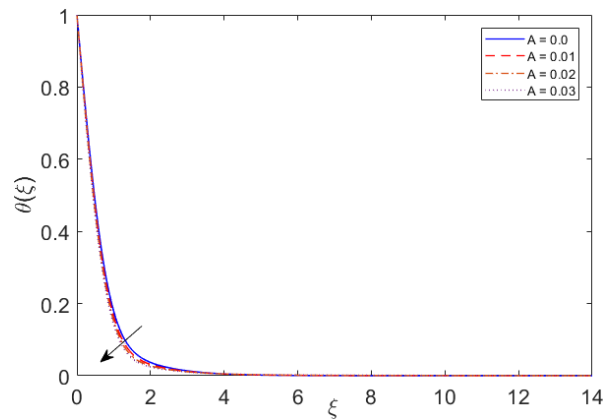


Figure 10. Temperature for varying A

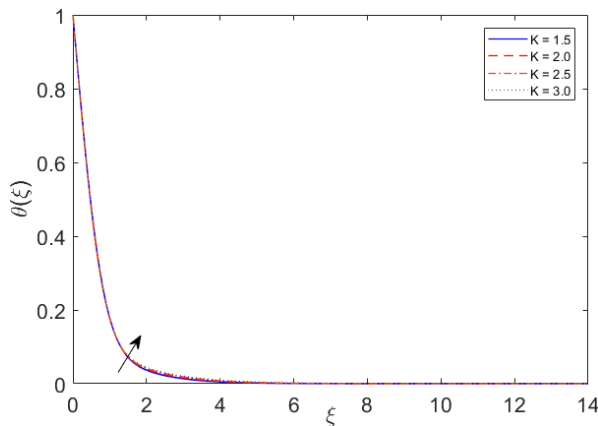


Figure 11. Temperature for varying K

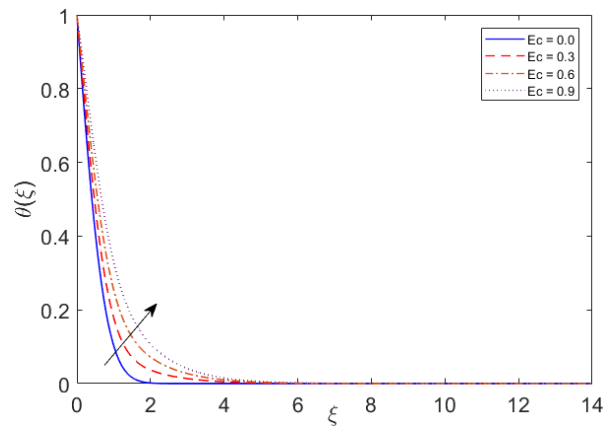


Figure 12. Temperature for varying Ec

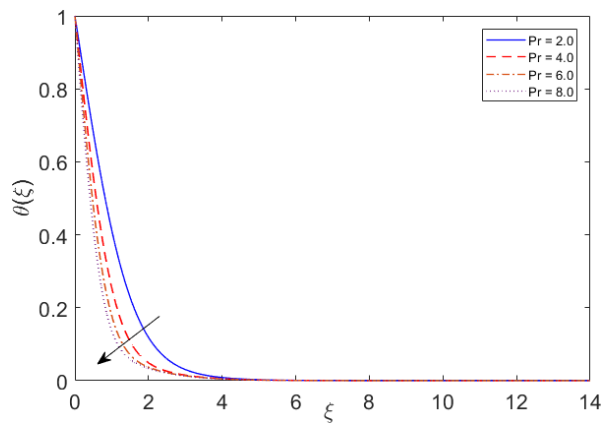


Figure 13. Temperature for varying Pr

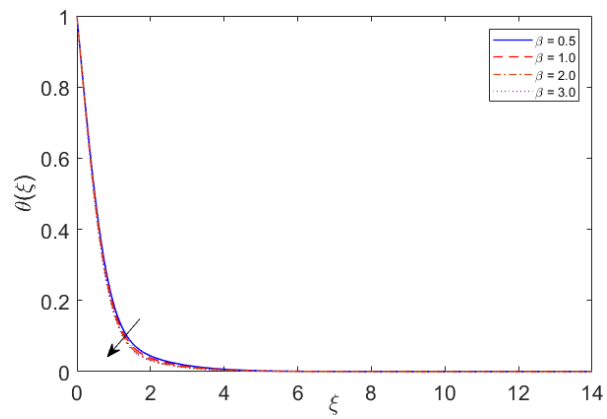


Figure 14. Temperature for varying β

5 Conclusions

This paper numerically examined the effects of MMR on the flow and heat transfer characteristics of a Williamson nanofluid over a stretching sheet. The governing equations were transformed into a system of ODEs and solved using the shooting method with Runge–Kutta integration. The results suggested that increasing the Williamson parameter suppressed the fluid velocity and reduced the momentum boundary-layer thickness due to enhanced non-Newtonian resistance. The magnetic parameter further decelerated the flow through the action of the Lorentz force, while the MMR parameter significantly altered the rotational behavior of fluid microelements. In addition, the temperature profile increased with the Eckert number because of viscous dissipation, whereas higher Prandtl numbers reduced the thermal boundary-layer thickness. The skin-friction coefficient and local Nusselt number were found to be highly sensitive to variations in the governing parameters. These findings demonstrated that MMR provided an additional mechanism for controlling momentum, microrotation, and heat transfer in magnetically responsive Williamson nanofluids. Such control may be advantageous in applications involving thermal management, magnetic cooling, and microfluidic transport systems. The present model was limited by the assumptions of steady two-dimensional flow and constant thermophysical properties. Future studies may consider unsteady and three-dimensional configurations, variable material properties, and more complex magnetic field effects to further improve the physical realism and engineering applicability of the model.

Author Contributions

Conceptualization, M.S.K.; methodology, E.H.A.; formal analysis, E.H.A.; investigation, M.S.K. and E.H.A.; data curation, M.S.K. and E.H.A.; writing—original draft preparation, M.S.K. and E.H.A.; writing—review and editing, M.S.K. and E.H.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

Data generated or analyzed during this study are provided in full within the published article.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

$\vec{V}$	Velocity vector, m/s
ρ_{nf}	Nanofluid density, kg/m ³
t	Time, s
j	Current density, A/m ²
H	Applied magnetic field, Wb/m ²
M_0	Magnetization strength, A/m
H_0	Magnetic field strength, A/m
$(\rho c_p)_{\text{nf}}$	Heat capacity, J/K
B	Magnetic induction vector, Wb/m ²
η_{nf}	Shear viscosity, kg/ms
η_1	Vortex viscosity, m ² /s
R_m	Relaxation magnetization factor, –
T	Fluid temperature, K
γ	Angular viscosity coefficient, rad/s
M	Magnetic field parameter, Wb/m ²
p	Pressure, N/m ²
Ec	Eckert number, –
β	Material parameter, –
ω^*	MMR parameter, –
K	Micropolar parameter, –
μ_0	Magnetic permeability, Tm/A
B_0	Magnetic induction strength, Wb/m ²
τ	Magnetization time relation, A/ms
Γ	Williamson material time constant, kg m s ^{−1}
Pr	Prandtl number, –
W	Microrotation vector, rad/s

Abbreviations

MMR	Micromagnetorotation
MHD	Magnetohydrodynamics